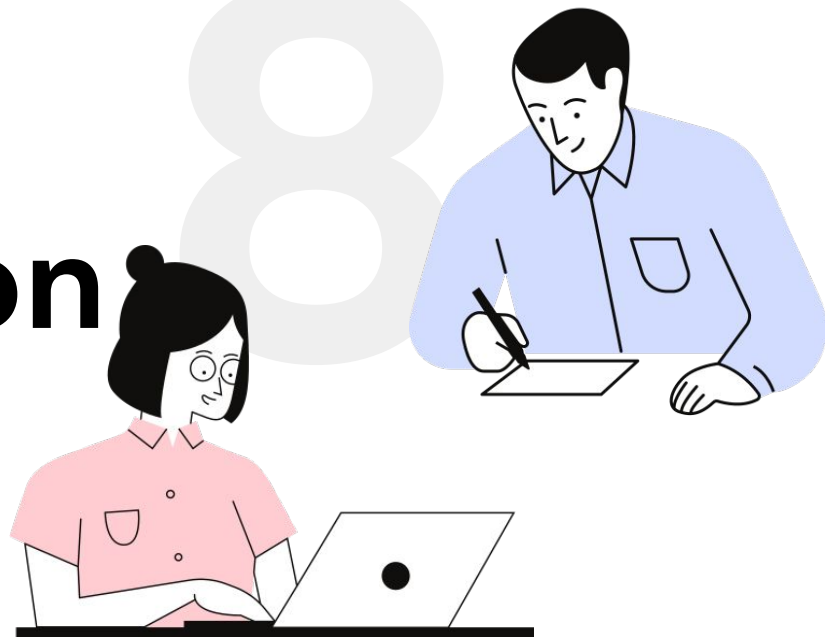




BRITISH INTERNATIONAL  
SCHOOL OF SULAYMANIYAH

January  
**2025**

# Information & Communication Technology



- ❖ *Identify and list common physical and e-safety risks associated with ICT usage.*
- ❖ *Explain the differences between various types of malware (viruses, worms, Trojan horses, spyware, adware, ransomware).*
- ❖ *Demonstrate how strong passwords and two-factor authentication can effectively secure online accounts.*
- ❖ *Compare and contrast biometric authentication methods, evaluating their advantages and disadvantages.*
- ❖ *Assess the effectiveness of encryption, digital certificates, SSL, and firewalls in maintaining data security.*



## Safety Issues

- Physical safety differs from health safety:
- Health safety prevents illness from daily computer use.
- Physical safety prevents serious injury or death from equipment hazards.
- Physical Safety Hazards & Prevention:

1. **Electrocution**
2. **Fire Hazard**
3. **Tripping Hazard**
4. **Personal Injury**

## Electrocution

### Causes

- Spilling liquids on electrical equipment
- Exposed wires/damaged insulation
- Unsafe electrical equipment or sockets

### Prevention

- No drinks near computers
- Regularly inspect and replace damaged wires
- Regular electrical inspections by qualified electricians
- Use Residual Current Breakers (RCBs)

## Fire Hazard

### Causes

- Overloaded sockets
- Overheating computer equipment
- Short circuits from exposed wires

### Prevention

- Increase available wall sockets; avoid extension overloads
- Do not block cooling vents
- Regularly clean computers of dust
- Regular equipment testing
- Ensure good room ventilation
- Prefer low-voltage equipment
- Maintain CO<sub>2</sub> or dry powder fire extinguishers

## Tripping Hazard

### Causes

- Trailing wires on floors
- Damaged flooring/carpets

### Prevention

- Use cable ducts; secure wires under desks
- Opt for wireless connections where possible

## Personal Injury

### Causes

- Unstable/falling heavy equipment
- Weak desks collapsing

### Prevention

- Use sturdy, adequately-sized desks
- Position equipment away from desk edges



## S25\_QP12\_Q5

The use of computer equipment can lead to many different safety issues.

For each of the following safety issues, **describe** three different ways of minimising the risk.

**(a)**      Electrocution

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**(b)**      Trailing leads

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| Question | Answer                                                                                                                                                                                                           | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(a)     | <b>Three</b> from:<br>Use Residual Circuit Breakers<br>Regularly check insulation on wires<br>Do not allow drinks near computer equipment<br>Regularly check equipment<br>Do not touch live wires with wet hands | <b>3</b> |
| 5(b)     | <b>Three</b> from:<br>Use cable ducts<br>Place cables under carpet / in false ceiling<br>Use cable management<br>Use wireless technology                                                                         | <b>3</b> |



- ❖ Data Protection Acts (DPAs) protect individuals from incorrect/inaccurate storage and processing of personal data.
- ❖ Applies to both computerised and paper records.
- ❖ Non-compliance can lead to fines or imprisonment.

## Principles of Data Protection Acts

- Fair and lawful processing.
- Data processed only for stated purposes.
- Adequate, relevant, and not excessive data.
- Accuracy.
- Data retention limited (not kept longer than needed).
- Compliance with individual's rights.
- Data security.
- No transfer to countries without adequate data protection.

## Prevent Unlawful Data Access

- ❖ Avoid leaving personal data unattended or visible.
- ❖ Lock filing cabinets and log off unattended computers.
- ❖ Securely use and frequently update passwords.
- ❖ Avoid sending sensitive information via email or fax.

- ❖ Personal data: Information that identifies a living person directly or indirectly.

## Examples of Personal Data

- Name, address, email, ID/passport numbers, IP addresses, cookie IDs, mobile ad identifiers, birthdate, banking details, photos.

## Sensitive Personal Data (Extra precautions needed):

- Ethnicity/race, political views, trade union membership, religion/philosophy, sexual orientation/gender, criminal records, medical/genetic/biometric data.
- Data confidentiality is crucial.
- Protect against hackers and accidental disclosure (encryption recommended).

- ❖ Covers internet, emails, social media, and online gaming.

## **Internet Safety Precautions**

- Trust secure websites (look for https and green padlock).
- Only purchase via encrypted connections.
- Activate "safe search" settings.
- Avoid harmful downloads; run updated anti-virus software.
- Log out properly, especially from sensitive sites.

## Email Safety Precautions

- Open emails/attachments only from known senders.
- Use ISP's spam filters.
- Verify sender addresses carefully (check for authenticity).
- Avoid including personal information in emails.
- Avoid sending identifiable photos (e.g., in uniform).
- Be wary of phishing/pharming scams.
- Regularly update passwords.
- Carefully handle forwarded emails.
- Manually type email addresses (avoid copying).
- Avoid clicking unknown links or using suspicious unsubscribe links.
- Use Bcc for group emails to protect privacy.



## Social Media Safety Precautions

- Don't publicly share personal details.
- Avoid posting identifiable images (e.g., uniform or car number plates).
- Strictly manage privacy settings.
- Only connect with trusted individuals.
- Report/block suspicious behavior or inappropriate language.

## **Safety in Chat Rooms and Instant Messaging**

- Always use nicknames, never real names.
- Keep private data confidential.
- Stay in public chat rooms.
- Never arrange private meetings alone; inform adults first.
- Respect others' confidentiality and avoid forwarding inappropriate images.

## **The rise of online gaming**

- Improved internet connections.
- More advanced mobile devices.
- Increased realism in games.

## **Risks associated with online gaming**

- Predators: targeting vulnerable users.
- Cyberbullying: threatening or intimidating messages.
- Webcam risks: potential misuse or inappropriate exposure.
- Voice-masking technology: used to conceal identity (age, gender, accent).
- Cyber attacks: including viruses, phishing, spyware.
- Exposure to violent content: may encourage real-life violence.

## **Safety Precautions in Online Gaming**

- Never reveal personal information, including real names.

## S25\_QP13\_Q10

Some personal data can be sensitive data

- (a) **Explain** the difference between personal data and sensitive data.

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- (b) State two examples of personal data.

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- (c) **State** two examples of sensitive data

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| Question | Answer                                                                                                                                                                                                              | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 10(a)    | <b>Two</b> from:<br>Personal data can be used to identify an individual<br>Personal data can be known by others<br>Sensitive data is protected by laws<br>Disclosure of sensitive data can be used against a person | 2     |
| 10(b)    | <b>Two</b> from, for example:<br>Address<br>Bank account number<br>Date of birth<br>Email address<br>Gender<br>Name<br>National Insurance number<br>Nationality<br>Telephone number                                 | 2     |
| 10(c)    | <b>Two</b> from, for example:<br>Criminal record<br>Medical history<br>Membership of a trade union<br>Political opinions<br>Racial origin<br>Religious beliefs<br>Sexual orientation                                | 2     |



Common security threats:

- ❖ Hacking
- ❖ Phishing, Vishing, Smishing
- ❖ Pharming
- ❖ Viruses and Malware
- ❖ Card fraud

## Hacking

Unauthorized access to computer systems.

- ❖ Possible Effects:
  - Identity theft.
  - Data deleted, changed, or corrupted.
- ❖ Prevention Methods:
  - Firewalls, anti-hacking software.
  - Strong, frequently changed passwords and user IDs.
  - Encryption (doesn't prevent hacking but makes data unreadable).

**Phishing:** Fake emails directing users to fraudulent websites or obtaining personal data.

**Smishing** (SMS Phishing): Fake SMS/text messages directing users to fraudulent websites or asking users to provide personal data via SMS or calls.

**Vishing** (Voicemail Phishing): Users tricked into calling fake telephone numbers, providing personal details.

## Possible Effects:

- Loss of sensitive personal data (credit card, bank details).
- Identity theft, fraud.
- Prevention Methods:
- Use ISP or browser filtering.
- Do not open unknown emails or suspicious attachments.
- Avoid clicking links or executable files (.exe, .bat, .com, .php).
- Verify authenticity (web/email addresses carefully).



- ❖ Malicious code redirects users unknowingly to fake websites.

## Possible Effects:

- Identity theft or financial fraud.

## Prevention Methods:

- Use anti-spyware software to detect/remove pharming code.
- Verify secure website indicators (https:// and padlock icon).

## Malware Types:

- Viruses
- Worms
- Trojan horse
- Adware
- Ransomware
- Keyloggers

## Viruses:

- ❖ Program code replicating itself, aiming to corrupt or delete files, causing system malfunction.

## Possible Effects:

- ❖ Computer crashes, becomes unresponsive.
- ❖ Data deletion, file corruption.

## ❖ Viruses prevention Methods:

- Regularly updated anti-virus software.
- Avoid software from unknown sources.
- Be cautious with email attachments from unknown senders.

## **Worms:**

- ❖ Stand-alone viruses that self-replicate and spread across networks without requiring user action.

## Possible Effects:

- Rapid spread throughout networks.
- Overloading networks, causing severe disruption (e.g., "I Love You" worm).

## Prevention Methods:

- Up-to-date anti-virus software.
- User caution with email attachments and network security maintenance.

- ❖ **Description:** Malicious software disguised as legitimate software.
- ❖ **Transmission:** Usually via email attachments or downloads from infected/fake websites.
- ❖ **Operation:** Requires user action (user runs or opens it).

## Effects:

- ❖ Gives cybercriminals access to personal data (IP addresses, passwords).
- ❖ Often installs spyware, ransomware, or key loggers.
- ❖ Risk Note: Firewalls and security software can be ineffective, as users override warnings.

- ❖ **Description:** Spyware monitoring user's keyboard activities.
- ❖ **Operation:** Records keystrokes and sends data (bank details, passwords) to cybercriminals.

## Prevention:

- Anti-spyware software for detection/removal.
- Banks request partial passwords or use dropdown boxes to prevent logging.
- Some key loggers take random screenshots (screen recorders).

**Description:** Malware flooding users with unwanted advertisements.

**Effects:**

- Redirects browsers to fake promotional websites.
- Displays pop-ups or unwanted browser toolbar changes.

**Risks:**

- Exposes weaknesses in user's security.
- Difficult to remove, as it's hard to distinguish harmful intent.
- Browser hijacking for fake search requests.

**Description:** Encrypts user data and demands payment ("ransom") to unlock.

**Transmission:** Via Trojan horses or social engineering.

**Effects:**

- Locks computer, making data inaccessible until payment.
- Severe impact on individuals/businesses.

**Prevention:**

- Regular backups (main method).
- Avoid opening suspicious emails or links.

Note: Once activated, reversing damage without backup is difficult or impossible.

**Viruses:** Self-replicating, require host program; corrupt/delete files.

**Worms:** Self-replicating without host; spread across networks.

**Trojan Horses:** Malicious programs disguised as legitimate software.

**Spyware:** Software monitoring computer activity to steal personal data.

**Adware:** Unwanted advertising software causing browser redirects/pop-ups.

**Ransomware:** Encrypts data, demands payment for data restoration.



Illegal use of credit/debit cards through various methods:

Shoulder surfing, Card cloning, Keylogging

## Shoulder Surfing

- ❖ Stealing data by observing users entering PINs or card details.
- ❖ Methods:
  - Observing directly, using cameras/binoculars.
  - Overhearing phone transactions.
  - Tiny hidden cameras near keypads.
- ❖ Prevention:
  - Shield keypad when entering data.
  - Avoid entering sensitive data publicly.
  - Use biometrics for enhanced security.

- ❖ **Description:** Copying card data onto fake cards using devices called skimmers.
- ❖ **Skimmers:** Inserted into ATMs; copy magnetic stripe data for counterfeit cards.
- ❖ **Shimmers:** Thin devices used on smart-chip cards to clone data onto fake magnetic-stripe cards.

## Prevention:

- Regularly monitor bank statements for unusual activity.
- Smart cards (chips) provide better security but are still vulnerable via shimmers.

## Key Logging

- Logs sensitive data (card number, PIN) for fraudulent online transactions.
- Use secure anti-spyware software and vigilance during online transactions to prevent it.

**Authentication Methods:** Ensure data is from secure/trusted sources:

- Biometrics
- Digital certificates
- Secure sockets layer (SSL)
- Encryption
- Firewalls
- Two-factor authentication
- User IDs and passwords

Uses unique human characteristics for security

- Fingerprint scans
- Signature recognition
- Retina/Iris scans
- Face recognition
- Voice recognition

## Fingerprint Scans

- Operation: Compares scanned fingerprint to database match.
- Applications: Secure entry to buildings, mobile device unlocking.
- ❖ Advantages:
  - Unique and hard to replicate.
  - Can't be lost/stolen/misplaced.
  - Prevents fraudulent "sign-ins".
- ❖ Disadvantages:
  - Expensive setup.
  - Injuries affecting fingers can reduce accuracy.
  - Concerns about civil liberties/privacy issues.

## Fingerprint scans

- ❖ Advantages:
  - High accuracy
  - Easy to use
  - Small storage requirement
- ❖ Disadvantages:
  - Intrusive perception (criminal association)
  - Errors if skin is dirty/damaged

## Face Recognition

- Description: Uses facial features to identify a user.
- Applications: Smartphones for unlocking and authorising purchases.
- Identification: distance between eyes, width of nose, length of jawline, ...etc.

### Advantages:

- Non-intrusive
- Relatively inexpensive

### Disadvantages:

- Easily affected by lighting changes, hairstyle/colour, ageing, or wearing spectacles.
- Privacy concerns due to storage of sensitive personal data.

## Iris recognition

- ❖ Advantages:
  - Highly accurate; fast verification (<5 sec)
- ❖ Disadvantages:
  - Very intrusive
  - Large data storage requirements
  - Expensive installation

## Voice recognition

- ❖ Advantages:
  - Non-intrusive; inexpensive
  - Fast verification
- ❖ Disadvantages:
  - Low accuracy, easily recorded
  - Voice changes due to illness



- ❖ Pair of files (public & private keys) ensuring secure data transmission.
- ❖ Confirms sender's identity and security of message content.

## Components:

- Sender's email address
- Owner's name
- Serial number
- Expiry date
- Public key (for encryption/signatures)
- Digital signature from Certificate Authority (CA), e.g., VeriSign.

- ❖ Protocol for securely sending/receiving data over the internet.
- ❖ Identified by: "https://" in URL & padlock symbol in browser
- ❖ Process:
  - Browser requests website connection.
  - Web server sends SSL certificate.
  - Browser authenticates certificate.
  - Encrypted two-way communication begins.
- ❖ Applications:
  - Online banking/shopping
  - Secure emails
  - Cloud storage
  - VoIP communications (audio/video calls)
  - Instant messaging/social networks

- ❖ Protects data from unauthorised access (makes data unreadable without key).
- ❖ Uses encryption key to convert plaintext into unreadable ciphertext; recipient uses decryption key to decode.
- ❖ Protects data on HDDs/SSDs.
- ❖ Encrypt connections to/from email providers.
- ❖ Encrypt actual messages and archives.
- ❖ Encrypt all emails (not just sensitive ones) to avoid targeting by hackers.
- ❖ Encrypt data before uploading to protect against breaches.

- ❖ Software/hardware monitoring network traffic; prevents unauthorised access.
  - ❖ Examines and filters incoming/outgoing data.
  - ❖ Blocks data failing security criteria.
  - ❖ Logs network traffic.
  - ❖ Prevents access to blacklisted IP addresses.
  - ❖ Alerts users about software accessing external sources.
- 
- ❖ Types:
    - Hardware (gateway devices)
    - Software (often part of the OS)

- ❖ Security method requiring two authentication steps to verify identity.
- ❖ Factors include:
  - Something known (password/PIN)
  - Something possessed (phone/tablet)
  - Something unique (biometrics)
- ❖ Example:
  - User logs in with username/password.
  - User receives a one-time passcode (OTP) via SMS/email.
  - User enters OTP to complete authentication.

- ❖ Passwords are used alongside User IDs to control and restrict access to data and computer systems.
- ❖ Strong password:
  - At least one capital letter
  - At least one number
  - At least one special keyboard character (e.g., @, \*, &)
- ❖ Difficult for others to guess but easy for the user to remember
  
- ❖ Example (strong password): Sy12@#TT90kj=0
- ❖ Example (weak password): GREEN
- ❖ Memory aid:
  - Phrase: “The 3rd planet is Earth: the 8th planet is Neptune”
  - Password: T3piE:t8piN (Strong and memorable)

## ❖ Protecting Passwords:

- Regularly run anti-spyware software to prevent password theft.
- Regularly change passwords to reduce security risks.
- Avoid easily guessable passwords, such as favourite colour, pet names, birthdays, etc.
- Ensure password fields show \*\*\*\*\* on-screen to prevent others from seeing.

## ❖ Password Verification and Security:

- Password entry often requires verification by typing twice (checks for input errors).
- Access is typically limited to a finite number of attempts (often three attempts), after which accounts are locked.
- Forgotten passwords usually require email verification and reset links, enhancing security by preventing unauthorised password changes.

S25\_QP12\_Q6

(b)

- (i) **State** two precautions that the user should take to protect their data from phishing attacks.

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- (ii) **State** two precautions that the user should take to protect their data from pharming attacks.

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| Question | Answer                                                                                                                                                                                                                                                                     | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 6(b)(i)  | <b>Two</b> from:<br>Use email filters<br>Check email addresses before opening emails<br>Do not reply to unsecured email addresses<br>Do not click on links within unsecured emails / untrusted websites<br>Do not give personal information in emails / untrusted websites | 2     |
| 6(b)(ii) | <b>Two</b> from:<br>Use up to date anti-malware to remove the malicious code<br>Check URL addresses<br>Do not give personal information to untrusted websites                                                                                                              | 2     |